

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

Product Name **1,4-Dioxane**  
 CAS No 123-91-1  
 Synonyms Diox

Product Code **96715**  
 Address ThermoFisher Scientific Australia Pty Ltd  
 5 Caribbean Drive, Scoresby  
 VICTORIA 3179, Australia  
 Emergency Tel. **CHEMTREC®**  
**03 9757 4559 or +613 9757 4559**  
 Telephone / Fax Numbers Tel: 1300 735 292  
 Fax: 1800 067 639  
 E-mail address ANZinfo@thermofisher.com

Recommended Use Laboratory chemicals.

Uses advised against This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.  
 This product does not contain any substance(s) subject to Prohibition, Authorization or  
 Restriction. This product does not contain any substance(s) listed on the voluntary National  
 Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

Flammable liquids

Category 2

#### Health hazards

Serious Eye Damage/Eye Irritation

Category 2

Carcinogenicity

Category 1B

Specific target organ toxicity - (single exposure)

Category 3

#### Environmental hazards

No hazards identified

### Label Elements



Flame



Exclamation Mark



Health Hazard

**Signal Word****Danger****Hazard Statements**

H225 - Highly flammable liquid and vapor  
H335 - May cause respiratory irritation  
H319 - Causes serious eye irritation  
H350 - May cause cancer  
AUH019 - May form explosive peroxides  
AUH066 - Repeated exposure may cause skin dryness or cracking

**Precautionary Statements**

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P233 - Keep container tightly closed  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P280 - Wear eye protection/ face protection  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P308 + P313 - IF exposed or concerned: Get medical advice/attention  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

Toxic to terrestrial vertebrates  
Contains a known or suspected endocrine disruptor  
Included in the list established in accordance with Article 59(1) for having endocrine disrupting properties

## Section 3 - Composition and Information on Ingredients

| Component   | CAS No   | Weight % |
|-------------|----------|----------|
| 1,4-Dioxane | 123-91-1 | >95      |

## Section 4 - First Aid Measures

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if

|                                            |                                                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | symptoms occur.                                                                                                                                  |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water.                                                                                     |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                                           |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                             |
| <b>Most important symptoms and effects</b> | . Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting                            |
| <b>Notes to Physician</b>                  | Treat symptomatically. Symptoms may be delayed.                                                                                                  |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

No information available.

### Hazardous Decomposition Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), peroxides.

### Specific Hazards Arising from the Chemical

Flammable. Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. May form explosive peroxides. Vapors may form explosive mixtures with air.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental Precautions

Should not be released into the environment.

### Methods for Containment and Clean Up

#### Clean-up methods - small spillage

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

#### Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if

available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

#### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

#### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges. If peroxide formation is suspected, do not open or move container. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools.

#### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Store under an inert atmosphere. Flammables area. May form explosive peroxides. Containers should be dated when opened and tested periodically for the presence of peroxides. Should crystals form in a peroxidizable liquid, peroxidation may have occurred and the product should be considered extremely dangerous. In this instance, the container should only be opened remotely by professionals. Keep away from heat, sparks and flame. Protect from moisture.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

#### Exposure limits

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)]

updated in August, 2005. Safe Work Australia **ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace. **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component   | Australia                                | New Zealand WEL                                 | ACGIH TLV           | The United Kingdom                                                                                                      | Germany                                                                                                                                                                                                                                                          |
|-------------|------------------------------------------|-------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1,4-Dioxane | TWA: 10 ppm<br>TWA: 36 mg/m <sup>3</sup> | TWA: 5 ppm<br>TWA: 18 mg/m <sup>3</sup><br>Skin | TWA: 20 ppm<br>Skin | STEL: 60 ppm 15 min<br>STEL: 219 mg/m <sup>3</sup> 15 min<br>TWA: 20 ppm 8 hr<br>TWA: 73 mg/m <sup>3</sup> 8 hr<br>Skin | TWA: 20 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 73 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 10 ppm (8 Stunden). MAK<br>TWA: 37 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 20 ppm<br>Höhepunkt: 74 mg/m <sup>3</sup><br>Haut |

#### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

| Component   | Australia | New Zealand | European Union | United Kingdom | Germany                                                                 |
|-------------|-----------|-------------|----------------|----------------|-------------------------------------------------------------------------|
| 1,4-Dioxane |           |             |                |                | 2-Hydroxyethoxyacetic acid: 200 mg/g<br>Creatinine urine (end of shift) |

#### Exposure Controls

##### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. Ensure that

eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### Personal protective equipment

##### Eye Protection

Tight sealing safety goggles Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

##### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments                                                                                                               |
|----------------|-------------------|-----------------|-----------------|------------------------------------------------------------------------------------------------------------------------------|
| Butyl rubber   | > 480 minutes     | 0.7 mm          | AS/NZS 2161     | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals<br>Permeation rate 38 µg/cm <sup>2</sup> /min |
| Viton (R)      | > 480 minutes     | 0.7 mm          |                 |                                                                                                                              |
| Butyl rubber   | < 200 minutes     | 0.35 mm         |                 |                                                                                                                              |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

##### Skin and body protection

Long sleeved clothing

##### Respiratory Protection

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

##### Recommended Filter type:

Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)

##### Recommended half mask:-

Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

#### Environmental exposure controls

No information available.

## Section 9 - Physical and Chemical Properties

#### Information on basic physical and chemical properties

|                            |                                             |                                          |
|----------------------------|---------------------------------------------|------------------------------------------|
| Appearance                 | Colorless                                   |                                          |
| Physical State             | Liquid                                      |                                          |
| Odor                       | Petroleum distillates                       |                                          |
| Odor Threshold             | No data available                           |                                          |
| pH                         | 6-8                                         | 500 g/l aq.sol                           |
| Melting Point/Range        | 12 °C / 53.6 °F                             |                                          |
| Softening Point            | No data available                           |                                          |
| Boiling Point/Range        | 101 °C / 213.8 °F                           | @ 760 mmHg                               |
| Flash Point                | 12 °C / 53.6 °F                             | <b>Method -</b> No information available |
| Evaporation Rate           | No data available                           |                                          |
| Flammability (solid,gas)   | Not applicable                              | Liquid                                   |
| Explosion Limits           | <b>Lower</b> 2 vol%<br><b>Upper</b> 22 vol% |                                          |
| Vapor Pressure             | 41 mbar @ 20 °C                             |                                          |
| Vapor Density              | 3                                           | (Air = 1.0)                              |
| Specific Gravity / Density | 1.034                                       |                                          |
| Bulk Density               | Not applicable                              | Liquid                                   |

|                                         |                          |                                             |
|-----------------------------------------|--------------------------|---------------------------------------------|
| Water Solubility                        | Soluble                  |                                             |
| Solubility in other solvents            | No information available |                                             |
| Partition Coefficient (n-octanol/water) |                          |                                             |
| Component                               | log Pow                  |                                             |
| 1,4-Dioxane                             | -0.42                    |                                             |
| Autoignition Temperature                | 355 °C / 671 °F          |                                             |
| Decomposition Temperature               | No data available        |                                             |
| Viscosity                               | 1.32 mPa.s @ 20 °C       |                                             |
| Explosive Properties                    |                          | Vapors may form explosive mixtures with air |
| Oxidizing Properties                    | No information available |                                             |
| <b>Other information</b>                |                          |                                             |
| Molecular Formula                       | C4 H8 O2                 |                                             |
| Molecular Weight                        | 88.11                    |                                             |

## Section 10 - Stability and Reactivity

|                                  |                                                                                                                                                                                                       |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reactivity                       | None known, based on information available                                                                                                                                                            |
| Stability                        | May form explosive peroxides. Hygroscopic.                                                                                                                                                            |
| Conditions to Avoid              | Incompatible products, Heat, flames and sparks, Exposure to air or moisture over prolonged periods, Keep away from open flames, hot surfaces and sources of ignition, Exposure to moist air or water. |
| Incompatible Materials           | Strong oxidizing agents, Reducing Agent, Halogens.                                                                                                                                                    |
| Hazardous Decomposition Products | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). peroxides.                                                                                                                                   |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                                                                                                              |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

|                     |                                                                  |
|---------------------|------------------------------------------------------------------|
| (a) acute toxicity; |                                                                  |
| Oral                | Based on available data, the classification criteria are not met |
| Dermal              | Based on available data, the classification criteria are not met |
| Inhalation          | Based on available data, the classification criteria are not met |

| Component   | LD50 Oral                                | LD50 Dermal                  | LC50 Inhalation       |
|-------------|------------------------------------------|------------------------------|-----------------------|
| 1,4-Dioxane | 5170 mg/kg ( Rat )<br>4200 mg/kg ( Rat ) | LD50 = 7600 mg/kg ( Rabbit ) | 48.5 mg/L ( Rat ) 4 h |

|                                |                                                                  |
|--------------------------------|------------------------------------------------------------------|
| (b) skin corrosion/irritation; | Based on available data, the classification criteria are not met |
|--------------------------------|------------------------------------------------------------------|

|                                    |            |
|------------------------------------|------------|
| (c) serious eye damage/irritation; | Category 2 |
|------------------------------------|------------|

|                                        |                                                                  |
|----------------------------------------|------------------------------------------------------------------|
| (d) respiratory or skin sensitization; |                                                                  |
| Respiratory                            | Based on available data, the classification criteria are not met |
| Skin                                   | Based on available data, the classification criteria are not met |

|                             |                                                                  |
|-----------------------------|------------------------------------------------------------------|
| (e) germ cell mutagenicity; | Based on available data, the classification criteria are not met |
|-----------------------------|------------------------------------------------------------------|

(f) carcinogenicity; Category 1B

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component   | Australia | New Zealand          | New South Wales | Western Australia | IARC     | EU           | UK | Germany |
|-------------|-----------|----------------------|-----------------|-------------------|----------|--------------|----|---------|
| 1,4-Dioxane |           | Confirmed carcinogen |                 |                   | Group 2B | Carc Cat. 1B |    |         |

(g) reproductive toxicity; Based on available data, the classification criteria are not met

(h) STOT-single exposure; Category 3

Results / Target organs Respiratory system

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Target Organs None known.

(j) aspiration hazard; Based on available data, the classification criteria are not met

Symptoms / effects, both acute and delayed Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

## Section 12 - Ecological Information

### Ecotoxicity effects

| Component   | Freshwater Fish                                                                                                                                                                                                                                                                              | Water Flea          | Freshwater Algae | Microtox                                                                  |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------|---------------------------------------------------------------------------|
| 1,4-Dioxane | LC50: > 10000 mg/L, 96h static (Lepomis macrochirus)<br>LC50: = 9850 mg/L, 96h (Pimephales promelas)<br>LC50: 10306 - 14742 mg/L, 96h static (Pimephales promelas)<br>LC50: = 9850 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: > 10000 mg/L, 96h semi-static (Lepomis macrochirus) | EC50 = 163 mg/L 48h |                  | EC50 = 610 mg/L 5 min<br>EC50 = 668 mg/L 15 min<br>EC50 = 733 mg/L 30 min |

Persistence and Degradability Not readily biodegradable  
 Persistence Persistence is unlikely.  
 Bioaccumulative Potential Bioaccumulation is unlikely

| Component   | log Pow | Bioconcentration factor (BCF) |
|-------------|---------|-------------------------------|
| 1,4-Dioxane | -0.42   | 0.3 - 0.7 dimensionless       |

Mobility The product is water soluble, and may spread in water systems. : Will likely be mobile in the environment due to its water solubility Highly mobile in soils

### Endocrine Disruptor Information

Persistent Organic Pollutant This product does not contain any known or suspected substance

Ozone Depletion Potential This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

Waste from Residues/Unused Products Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

|                               |                                                                                                                                                                                                                                                                                                    |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Contaminated Packaging</b> | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                           |
| <b>Other Information</b>      | Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. |

## Section 14 - Transport Information

### IMDG/IMO

|                             |         |
|-----------------------------|---------|
| <b>UN-No</b>                | UN1165  |
| <b>Proper Shipping Name</b> | DIOXANE |
| <b>Hazard Class</b>         | 3       |
| <b>Packing Group</b>        | II      |

### ADG

|                             |         |
|-----------------------------|---------|
| <b>UN-No</b>                | UN1165  |
| <b>Proper Shipping Name</b> | DIOXANE |
| <b>Hazard Class</b>         | 3       |
| <b>Packing Group</b>        | II      |

| Component                       | Hazchem Code |
|---------------------------------|--------------|
| 1,4-Dioxane<br>123-91-1 ( >95 ) | 2YE          |

### IATA

|                             |         |
|-----------------------------|---------|
| <b>UN-No</b>                | UN1165  |
| <b>Proper Shipping Name</b> | DIOXANE |
| <b>Hazard Class</b>         | 3       |
| <b>Packing Group</b>        | II      |

|                               |                                 |
|-------------------------------|---------------------------------|
| <b>Environmental hazards</b>  | No hazards identified           |
| <b>Special Precautions</b>    | No special precautions required |
| <b>Additional information</b> | None known                      |

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations      **Australia**

See section 8 for national exposure control parameters.

#### **Standard for the Uniform Scheduling of Medicines and Poisons**

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

| Component              | Standard for the Uniform Scheduling of Medicines and Poisons |
|------------------------|--------------------------------------------------------------|
| 1,4-Dioxane - 123-91-1 | Schedule 6 listed - present                                  |

#### **Australian Industrial Chemicals Introduction Scheme (AICIS)**



| Component              | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information                                                                                                                                                                                                       |
|------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1,4-Dioxane - 123-91-1 | Present                                                     | Specific information requirement: Obligations to provide information apply. You must tell us within 28 days if the circumstances of your importation or manufacture (introduction) are different to those in our assessment. |

**Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

**Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory** Not applicable

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

| Component              | Australia | New South Wales | Western Australia | New Zealand          |
|------------------------|-----------|-----------------|-------------------|----------------------|
| 1,4-Dioxane - 123-91-1 |           |                 |                   | Confirmed carcinogen |

**International Inventories**

| Component   | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|-------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| 1,4-Dioxane | X    | X     | 204-661-8 | -      | X    | X   | -    | X     | X    | X    | X     | KE-10463 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**

Not applicable.

| Component   | CAS No   | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-------------|----------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 1,4-Dioxane | 123-91-1 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

**Authorisation/Restrictions according to EU REACH**

| Component   | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)                                                                                                                                                                                          |
|-------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1,4-Dioxane | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)<br>Use restricted. See item 28.<br>(see link for restriction details) | SVHC Candidate list - 204-661-8 - Carcinogenic (Article 57a)<br><br>Equivalent level of concern having probable serious effects to the environment (Article 57f - environment)<br><br>Equivalent level of concern having probable serious effects to human health (Article 57f - human health) |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

<https://echa.europa.eu/authorisation-list>

## Section 16 - Other Information

### Legend

**AICS** - Australian Inventory of Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**NZS 5433:2012** - Transport of Dangerous Goods on Land

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**NZIoC** - New Zealand Inventory of Chemicals

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists  
Predicted No Effect Concentration (PNEC)

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**ADG** Australian Code for the Transport of Dangerous Goods by Road and Rail

**OECD** - Organisation for Economic Co-operation and Development

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

### Revision Date

18-Nov-2022

### Revision Summary

SDS sections updated.

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**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**